Logic and Sets for AI

Midterm prep
Mentor Program

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General info

- Week 1-3 of logic and sets
- On paper!

1. **Semantic equivalence** $\equiv$ (9 points)
   Group the following eight formulas into their semantic equivalence classes (i.e., state which formulas are semantically equivalent):

$$p \to (q \to r) \qquad (p \to q) \to r \qquad (p \to q) \leftrightarrow r \qquad (p \leftrightarrow q) \leftrightarrow r$$

$$p \leftrightarrow (q \leftrightarrow r) \qquad (p \wedge q) \to r \qquad p \oplus (q \oplus r) \qquad \neg p \vee \neg q \vee r$$

Solution:

The equivalence classes are:

- $p \to (q \to r) \quad (p \wedge q) \to r \quad \neg p \vee \neg q \vee r$
- $(p \leftrightarrow q) \leftrightarrow r \quad p \leftrightarrow (q \leftrightarrow r) \quad (p \oplus q) \oplus r$
- $(p \to q) \to r$
- $(p \to q) \leftrightarrow r$

2. Island puzzle (12 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet islanders A and B.

A says: "If I am a liar, then B is a truth speaker."

B says: "Exactly one of us two is a liar."

Determine via a truth table for both A and B whether this islander is a truth speaker or a liar.

Solution:

t_A	t_B	$\neg t_A \to t_B$	$t_A \oplus t_B$	$t_A \leftrightarrow (\neg t_A \to t_B)$	$t_B \leftrightarrow (t_A \oplus t_B)$
T	T				
T	F				
F	T				
F	F				

Solution:

t_A	t_B	$\neg t_A \to t_B$	$t_A \oplus t_B$	$t_A \leftrightarrow (\neg t_A \to t_B)$	$t_B \leftrightarrow (t_A \oplus t_B)$
T	T	T	F	T	F
T	F	T	T	T	F
F	T	T	T	F	T
F	F	F	F	T	T

Solution:

t_A	t_B	$\neg t_A \to t_B$	$t_A \oplus t_B$	$t_A \leftrightarrow (\neg t_A \to t_B)$	$t_B \leftrightarrow (t_A \oplus t_B)$
T	T	T	F	T	F
T	F	T	T	T	F
F	T	T	T	F	T
F	F	F	F	T	T

Only in the 4th row do the 5th and 6th column both have a T. So A and B are both liars.

3. Conjunctive normal form (12 points)

Apply the algorithm `CNF` to turn the formula

$$\neg((p \vee q) \rightarrow r) \vee (\neg p \wedge r)$$

into CNF.

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into CNF.

1. `IMPL-FREE`: Eliminate all occurrences of $\rightarrow$.

$$\phi \rightarrow \psi \equiv \neg\phi \vee \psi$$

2. `NNF`: Bring all occurrences of $\neg$ inside, until they a directly in front of a propositional variable.

$$\neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi \qquad \neg\neg\phi \equiv \phi$$

$$\neg(\phi \vee \psi) \equiv \neg\phi \wedge \neg\psi$$

3. `DISTR`: Use distributivity to re-arrange $\vee$ and $\wedge$.

$$(\phi \wedge \psi) \vee \chi \equiv (\phi \vee \chi) \wedge (\psi \vee \chi)$$

$$\chi \vee (\phi \wedge \psi) \equiv (\chi \vee \phi) \wedge (\chi \vee \psi)$$

Solution:

$$
\begin{aligned}
\neg((p \vee q) \rightarrow r) \vee (\neg p \wedge r) \quad & \rightsquigarrow_{\texttt{IMPL-FREE}} && \neg(\neg(p \vee q) \vee r) \vee (\neg p \wedge r) \\
& \rightsquigarrow_{\texttt{NNF}} && (\neg\neg(p \vee q) \wedge \neg r) \vee (\neg p \wedge r) \\
& \rightsquigarrow_{\texttt{NNF}} && ((p \vee q) \wedge \neg r) \vee (\neg p \wedge r) \\
& \rightsquigarrow_{\texttt{DISTR}} && ((p \vee q) \vee (\neg p \wedge r)) \wedge (\neg r \vee (\neg p \wedge r)) \\
& \rightsquigarrow_{\texttt{DISTR}} && (((p \vee q) \vee \neg p) \wedge ((p \vee q) \vee r)) \wedge (\neg r \vee (\neg p \wedge r)) \\
& \rightsquigarrow_{\texttt{DISTR}} && (((p \vee q) \vee \neg p) \wedge ((p \vee q) \vee r)) \wedge ((\neg r \vee \neg p) \wedge (\neg r \vee r))
\end{aligned}
$$

4. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee \neg r) \wedge (\neg p \vee r)$$

to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Let the procedure branch first on p, then on q, and then on r; and let it always try the truth value true ($\top$) before false ($\bot$).

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We check the satisfiability of a CNF ϕ.

- eliminate "unit" clauses:
  - for each *unit clause* p of ϕ, replace all p's in ϕ by $\top$
  - for each *unit clause* $\neg p$ of ϕ, replace all p's in ϕ by $\bot$
- eliminate "pure" propositional variables:
  - if a p occurs only *positively* in ϕ, replace all p's in ϕ by $\top$
  - if a p occurs only *negatively* in ϕ, replace all p's in ϕ by $\bot$

$$\psi \wedge \top \rightsquigarrow \psi \qquad \psi \vee \top \rightsquigarrow \top \qquad \neg\top \rightsquigarrow \bot$$
$$\top \wedge \psi \rightsquigarrow \psi \qquad \top \vee \psi \rightsquigarrow \top \qquad \neg\bot \rightsquigarrow \top$$
$$\psi \wedge \bot \rightsquigarrow \bot \qquad \psi \vee \bot \rightsquigarrow \psi$$
$$\bot \wedge \psi \rightsquigarrow \bot \qquad \bot \vee \psi \rightsquigarrow \psi$$

Solution: At the start there are neither unit clauses nor pure propositional variables.

In the first step, $\top$ is substituted for p, leading to

$$(\top \vee q) \wedge (\neg\top \vee \neg r) \wedge (\neg q \vee \neg r) \wedge (\neg\top \vee r)$$

In 5 steps, this formula reduces to $\neg r \wedge (\neg q \vee \neg r) \wedge r$.

In view of the unit clause r (or $\neg r$), $\top$ (or $\bot$) can be substituted to r. In both cases the resulting CNF contains a conjunct $\bot$, so that the overall formula reduces to $\bot$.

Now the algorithm backtracks and $\bot$ is substituted for p, leading to

$$(\bot \vee q) \wedge (\neg\bot \vee \neg r) \wedge (\neg q \vee \neg r) \wedge (\neg\bot \vee r)$$

In 7 steps, this formula reduces to $q \wedge (\neg q \vee \neg r)$.

In view of the unit clause q, $\top$ can be substituted for q. And inview of the (negative) pure variable r, $\bot$ can be substituted for r. This leads to

$$\top \wedge (\neg\top \vee \neg\bot)$$

In 3 steps, this formula reduces to $\top$.

Concluding, DPLL finds the following satisfaction of the original formula: $p = \bot$, $q = \top$ and $r = \bot$.

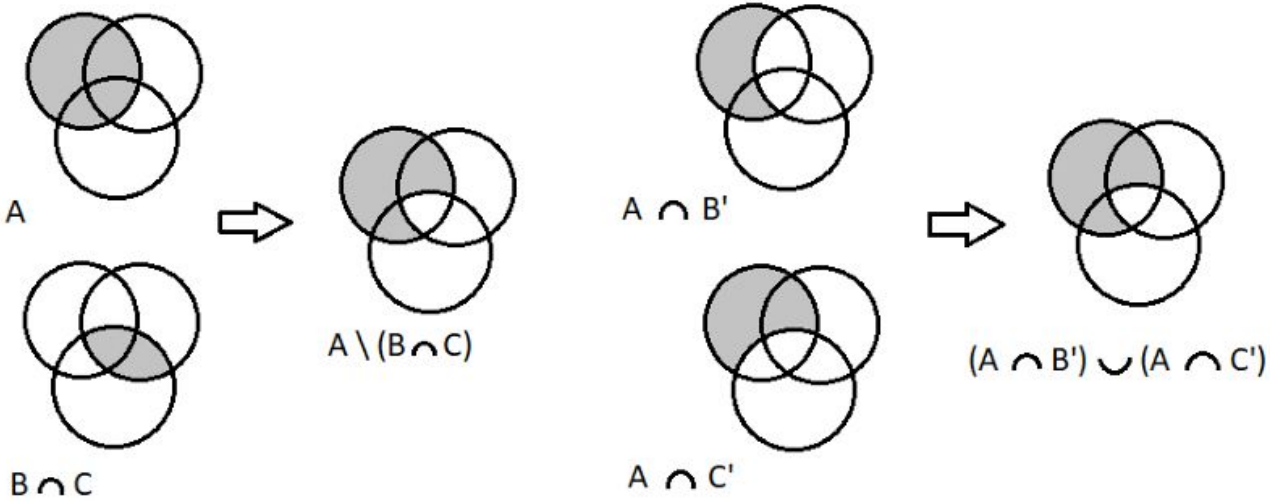
5. Sets (5+5+5 points) Consider the following set-theoretic equality

$$A \backslash (B \cap C) = (A \cap B') \cup (A \cap C').$$

a) Prove this identity using a sequence of Venn diagrams.

Solution:

a) Using Venn diagrams we see that



5. Sets (5+5+5 points) Consider the following set-theoretic equality

$$A \backslash (B \cap C) = (A \cap B') \cup (A \cap C').$$

a) Prove this identity using a sequence of Venn diagrams.

b) Prove it using the rules of the algebra of sets.

$$A \backslash (B \cap C) = (A \cap B') \cup (A \cap C').$$

Commutativity

$A \cap B = B \cap A$

$A \cup B = B \cup A$

Associativity

$A \cap (B \cap C) = (A \cap B) \cap C$

$A \cup (B \cup C) = (A \cup B) \cup C$

Complement

$A \cap A' = \varnothing$

$A \cup A' = U$

Identity

$A \cap U = A$

$A \cup \varnothing = A$

Domination

$A \cap \varnothing = \varnothing$

$A \cup U = U$

Involution

$(A')' = A$

Idempotence

$A \cap A = A$

$A \cup A = A$

Definitions

$A \setminus B = A \cap B'$

$A \triangle B = (A \cup B) \cap (A \cap B)'$

De Morgan

$(A \cap B)' = A' \cup B'$

$(A \cup B)' = A' \cap B'$

Distributivity

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

b)

$$
\begin{aligned}
A \backslash (B \cap C) &= A \cap (B \cap C)' && \text{(Definitions)} \\
&= A \cap (B' \cup C') && \text{(De Morgan)} \\
&= (A \cap B') \cup (A \cap C') && \text{(Distributivity)}
\end{aligned}
$$

5. Sets (5+5+5 points) Consider the following set-theoretic equality

$$A \backslash (B \cap C) = (A \cap B') \cup (A \cap C').$$

a) Prove this identity using a sequence of Venn diagrams.

b) Prove it using the rules of the algebra of sets.

c) Assuming that $\#A = 5$, $\#(B' \cup C') = 5$, $\#(A \cup B' \cup C') = 7$, compute the number of elements in $A \backslash (B \cap C)$.

c) Since

$$\#(A \cup B' \cup C') = \#A + \#(B' \cup C') - \#(A \cap (B' \cup C'))$$

we have

$$\begin{aligned} \#(A \backslash (B \cap C)) &= \#(A \cap (B' \cup C')) \\ &= \#A + \#(B' \cup C') - \#(A \cup B' \cup C') \\ &= 5 + 5 - 7 = 3. \end{aligned}$$

6. Relations (6+5+6 points) In the set $V = \{1, 2, 3, 4\}$ consider the relations

$$\begin{aligned} R &= \{\langle 1,2\rangle, \langle 1,3\rangle, \langle 1,4\rangle, \langle 2,2\rangle, \langle 2,3\rangle, \langle 2,4\rangle, \langle 3,3\rangle\}, \\ S &= \{\langle 1,1\rangle, \langle 2,3\rangle, \langle 3,4\rangle\}. \end{aligned}$$

a) Write down the matrix representation of the relation R.

Solution:

a) $$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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b) Is the relation R transitive? Is it reflexive? (Explain your answers.)

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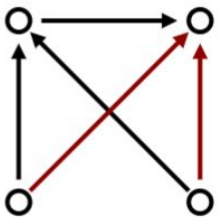
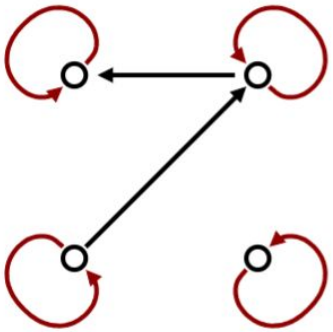
b) Is the relation R transitive? Is it reflexive? (Explain your answers.)

Transitivity



for all elements x, y, z:
$x\,R\,y \wedge y\,R\,z \rightarrow x\,R\,z$

Reflexivity



for all elements x:
$x\,R\,x$

b) – R is *not* reflexive, since $\langle 1,1\rangle \notin R$ *or* $\langle 4,4\rangle \notin R$

– R is transitive, since $\langle 1,2\rangle \in R \wedge \langle 2,3\rangle \in R \Rightarrow \langle 1,3\rangle \in R$ *and* $\langle 1,2\rangle \in R \wedge \langle 2,4\rangle \in R \Rightarrow \langle 1,4\rangle \in R$.

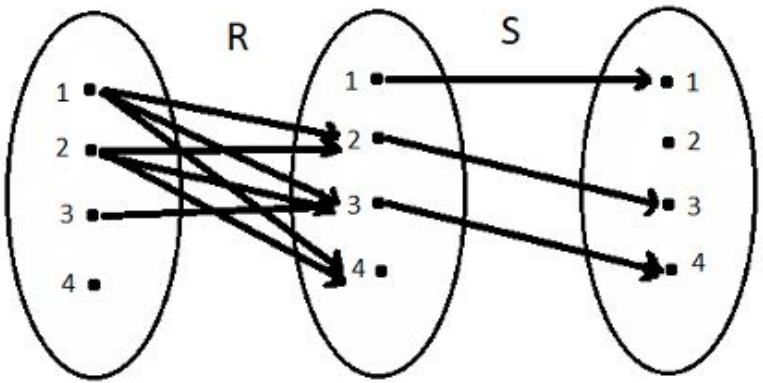
6. Relations (6+5+6 points) In the set $V = \{1, 2, 3, 4\}$ consider the relations

$$\begin{aligned} R &= \{\langle 1,2\rangle, \langle 1,3\rangle, \langle 1,4\rangle, \langle 2,2\rangle, \langle 2,3\rangle, \langle 2,4\rangle, \langle 3,3\rangle\}, \\ S &= \{\langle 1,1\rangle, \langle 2,3\rangle, \langle 3,4\rangle\}. \end{aligned}$$

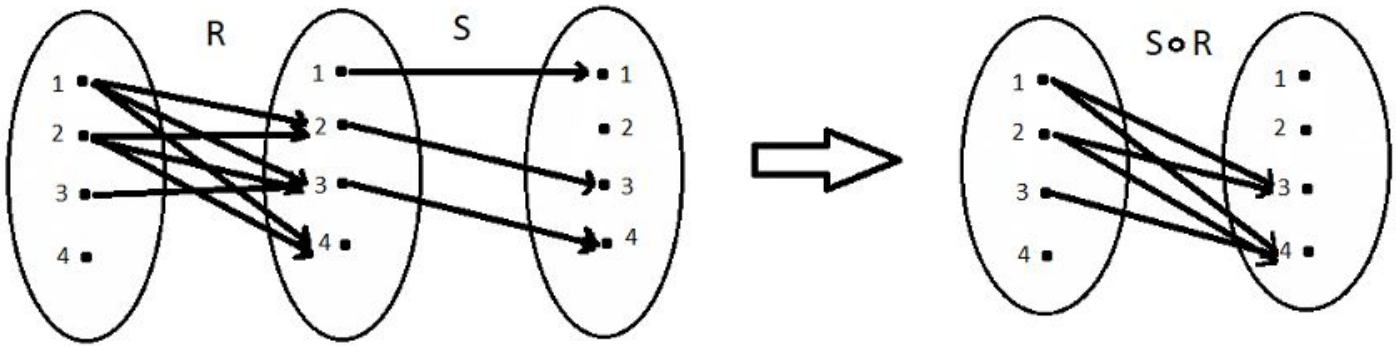
a) Write down the matrix representation of the relation R.

b) Is the relation R transitive? Is it reflexive? (Explain your answers.)

c) Draw a Venn diagram depicting the relation $S \circ R$.


R
S
1
2
3
4
1
2
3
4
1
2
3
4

c) We find $S \circ R = \{\langle 1,3\rangle, \langle 1,4\rangle, \langle 2,3\rangle, \langle 2,4\rangle, \langle 3,4\rangle\}$

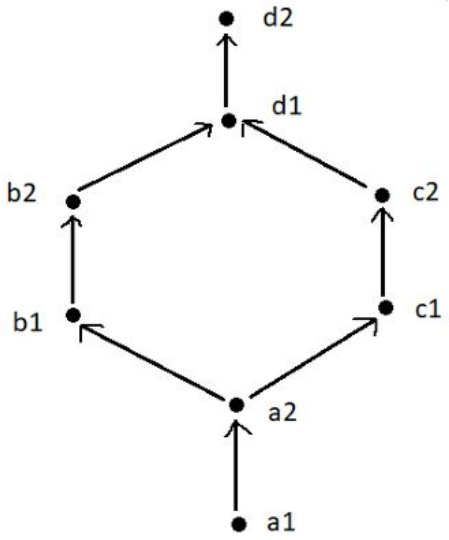


7. Ordering relations (7+6 points) Consider $A = \{1, 2\}$ and $B = \mathcal{P}(A)$, the power set of A, both equipped with their natural partial ordering: for A the relation $\leq$, and for B the relation given by set inclusion.

a) Draw the Hasse diagram of the lexicographic order of $B \times A$. Explain whether it is a total ordering.

Solution:

a) It is *not* a total ordering. Abbreviating $a = \{\}$, $b = \{1\}$, $c = \{2\}$ and $d = \{1, 2\}$ we find that the lexicographical ordering on $B \times A$ is given by

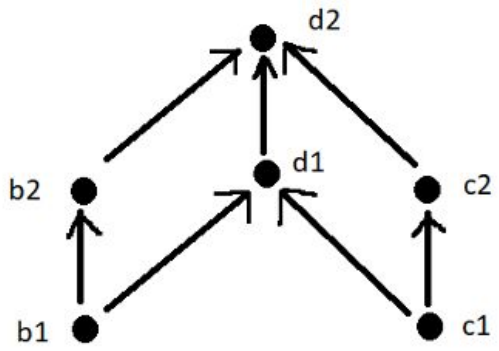


7. Ordering relations (7+6 points) Consider $A = \{1, 2\}$ and $B = \mathcal{P}(A)$, the power set of A, both equipped with their natural partial ordering: for A the relation $\leq$, and for B the relation given by set inclusion.

a) Draw the Hasse diagram of the lexicographic order of $B \times A$. Explain whether it is a total ordering.

b) Let C denote the set B with the empty set removed. List the minimal and the maximal elements of $C \times A$ equipped with the Cartesian order. Is there a largest or a smallest element of $C \times A$?

b) $\langle\{1\},1\rangle$ and $\langle\{2\},1\rangle$ are the minimal elements, $\langle\{1,2\},2\rangle$ is the only maximal element. The largest element is $\langle\{1,2\},2\rangle$ and there is no smallest element.



Final tips

- Work carefully
- Use the scrap paper
- Do the exercises + prior exams
- Be confident!

Thank You !